Calculus II HW 2

Part I. Regular problems

1. Determine convergence or divergence of the series.

(a) $\sum_{k=2}^{\infty} \frac{2}{k \ln k}$

(b) $\sum_{k=2}^{\infty} \frac{1}{\ln(k!)}$

(c) $\sum_{k=1}^{\infty} \frac{ke^{-k^2}}{4+e^{-k}}$

(d) $\sum_{k=1}^{\infty} \frac{\sin^{-1}(1/k)}{k^2}$

(e) $\sum_{k=1}^{\infty} \frac{\cos(k\pi/5)}{k!}$

2. (a) Show that $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^4}$ converges.

(b) Approximate the sum of the alternating series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^4}$ by the 40-th partial sum $S_{40} \approx 0.9470326439$ and estimate the error in this approximation.

(c) For the convergent alternating series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^4}$, how many terms are needed to guarantee that the partial sum S_n is within 10^{-10} of the actual sum?

3. Prove that for $a_k > 0$, $\sum_{k=1}^{\infty} a_k$ converges if and only if $\sum_{k=1}^{\infty} \frac{a_k}{1+a_k}$ converges.

(Hint: If $0 < x < 1$, then $x < \dfrac{2x}{1+x}$).

4. Determine whether the series is absolutely convergent, conditionally convergent or divergent

(a) $\displaystyle\sum_{k=5}^{\infty}\left(\frac{1-3k}{4k}\right)^k$

(b) $\displaystyle\sum_{k=1}^{\infty}k^3e^{-k}$

(c) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{e^k}{k^2}\right)^k$

(d) $\displaystyle\sum_{k=1}^{\infty}(-1)^{k+1}\frac{k^{10}}{(2k)!}$

(e) $\displaystyle\sum_{k=1}^{\infty}\frac{k!}{1\cdot3\cdot5\cdots(2k-1)}$

(f) $\displaystyle\sum_{k=4}^{\infty}(-1)^{k+1}\frac{k^2\cdot4^k}{k!}$

(g) $\displaystyle\sum_{k=2}^{\infty}(-1)^{k+1}\frac{k}{2k+1}$

Part II. Group problems

1.

Test the series for convergence or divergence.

(a) $-\frac{2}{5}+\frac{4}{6}-\frac{6}{7}+\frac{8}{8}-\frac{10}{9}+\cdots$

(b) $\sum_{n=0}^{\infty}\frac{(-1)^{n+1}}{\sqrt{n+1}}$

(c) $\sum_{n=1}^{\infty}(-1)^n\frac{n}{2^n}$

2. Determine whether the series is absolutely convergent, conditionally convergent or divergent.

(a) $\sum_{k=6}^{\infty}(-1)^k \frac{k \cdot 2^k}{3^k}$

(b) $\sum_{k=1}^{\infty}(-1)^{k+1} \frac{k}{k^2+1}$

(c) $\sum_{k=1}^{\infty}\frac{(-1)^k}{k \ln k}$